



Task 3: Multi-Objective Coordination

Efficiency vs Fairness in
Cooperative Mingle
*Pareto frontier exploration with
scalarized rewards*



Research Question & Hypothesis

**Research question:**

Can we map the Pareto frontier between coordination efficiency and fairness?

**Hypothesis:**

Higher efficiency weight should increase reward and room occupancy, while lower efficiency weight should support fairer participation.

Implementation

Scalarized reward:

$$R = \alpha * R_{efficiency} + (1 - \alpha) * R_{fairness}$$

Implemented components:

- EfficiencyReward
- FairnessReward
- MultiObjectiveReward
- fairness metrics: Gini, Jain fairness, participation range

Experiment Setup



Alpha values:

0.0, 0.25, 0.5, 0.75, 1.0



Small scenarios:

4/2, 4/3, 6/2, 6/3



Large scenarios:

8/4, 8/5, 10/4, 10/5

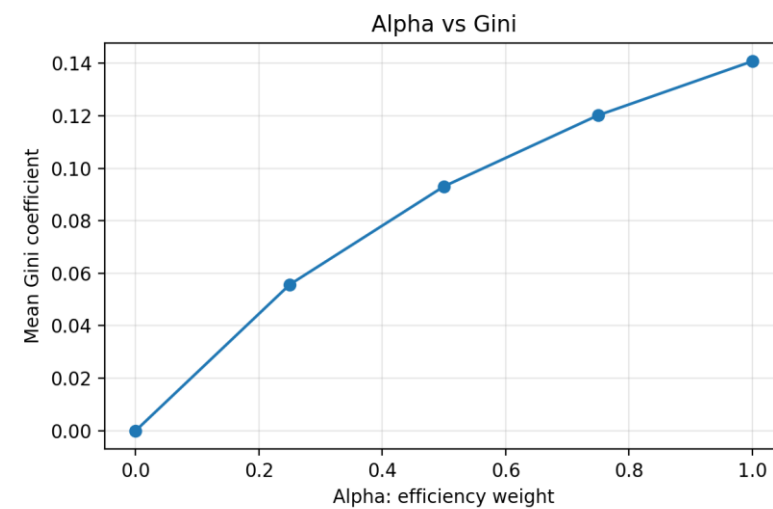
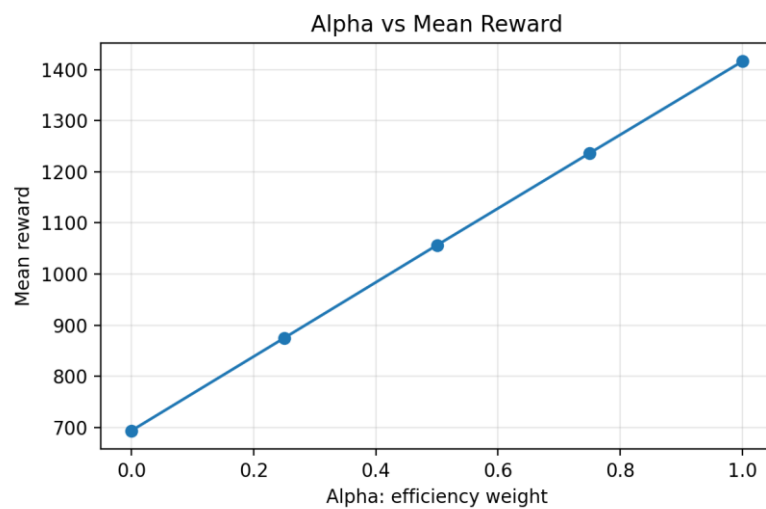
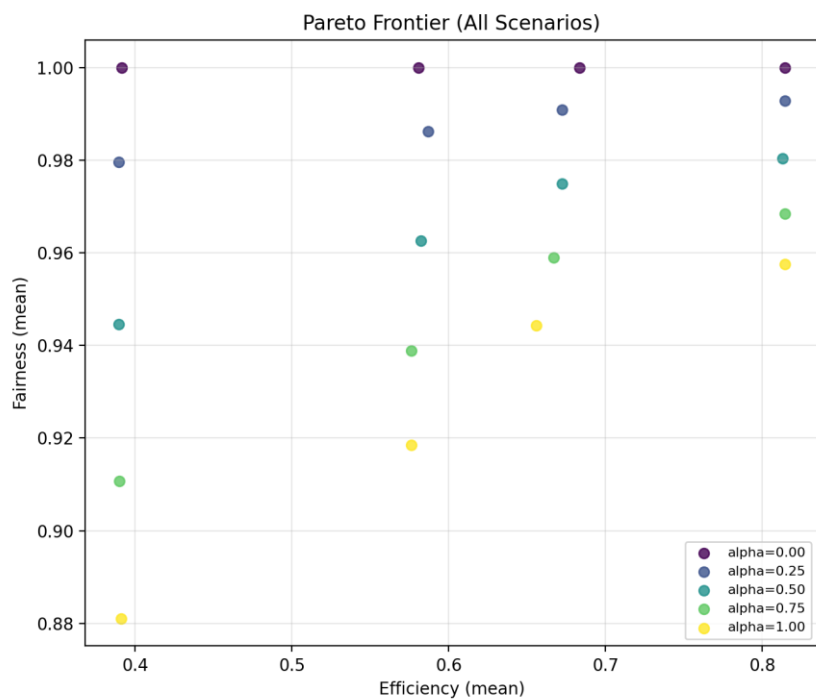


Seeds:

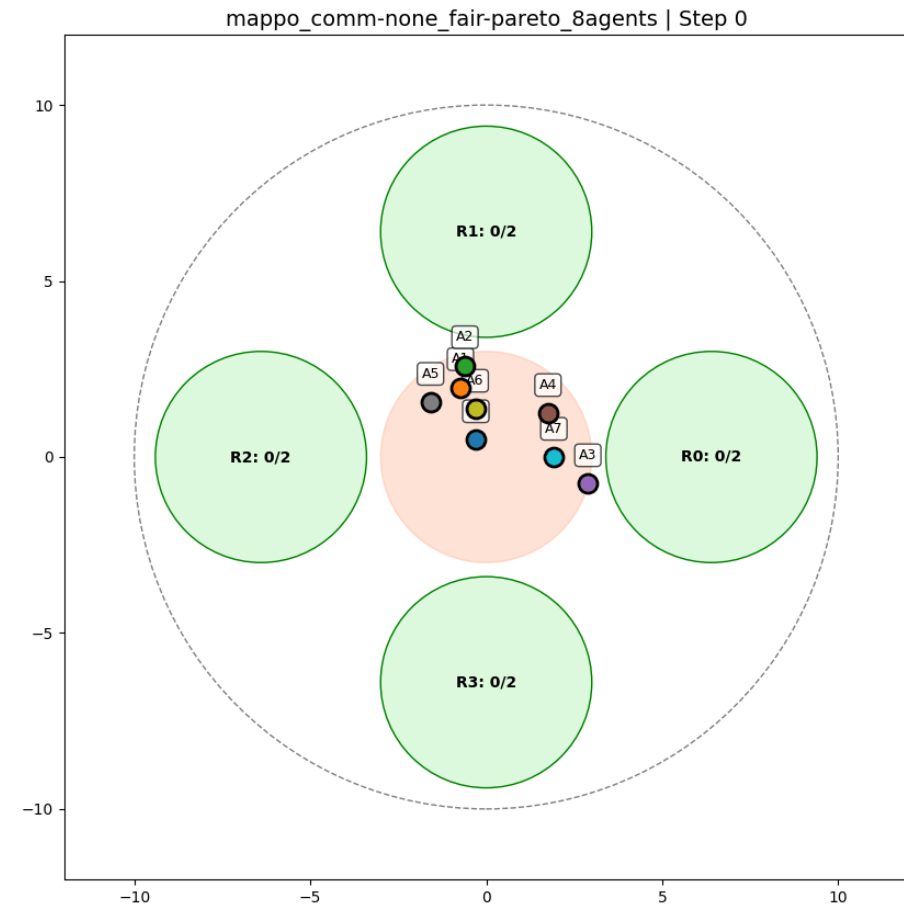
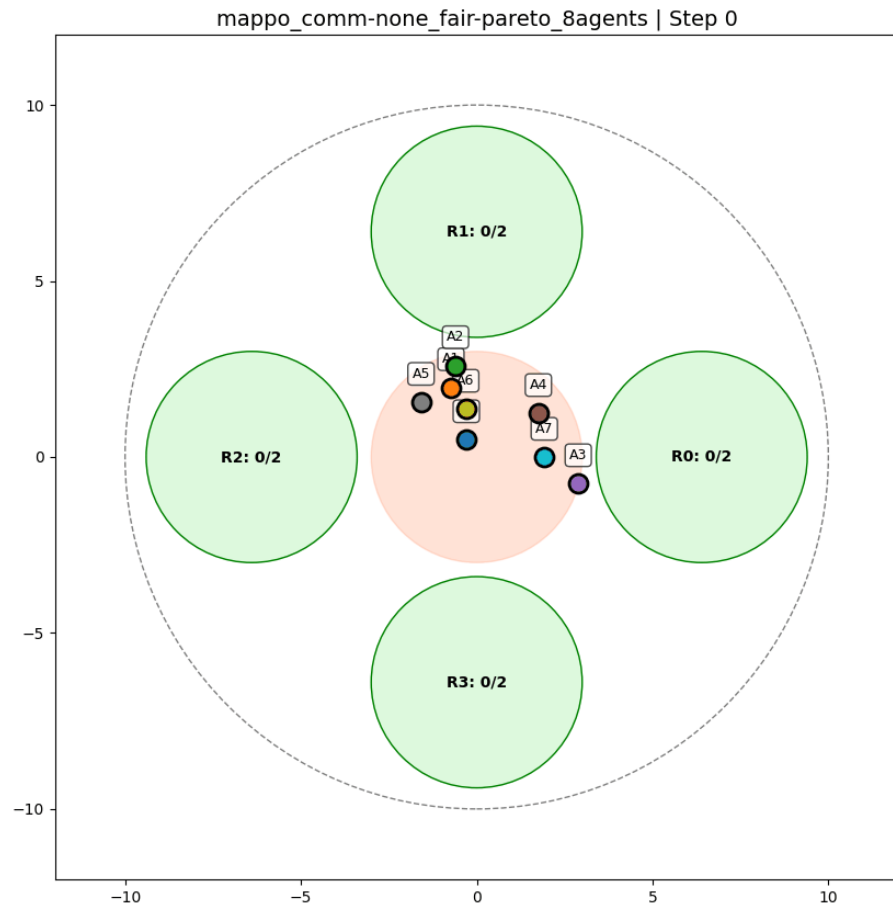
5 fixed seeds per
condition

one extra smoke run for
8/4 large

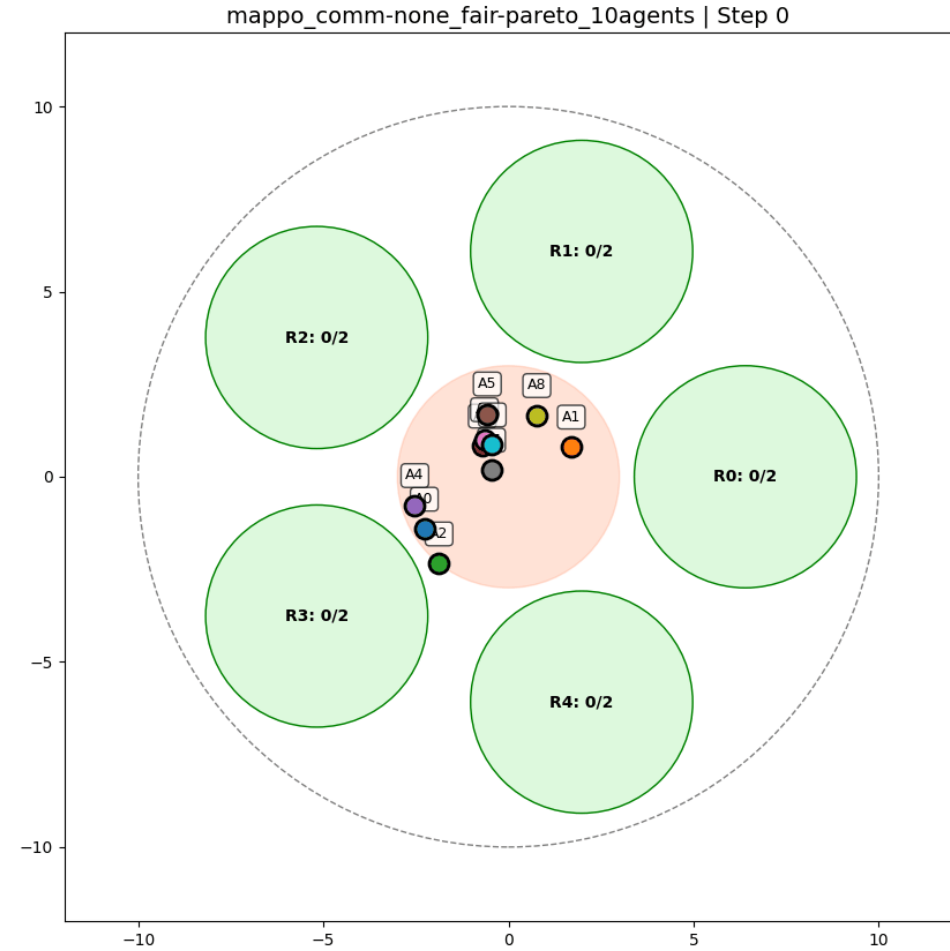
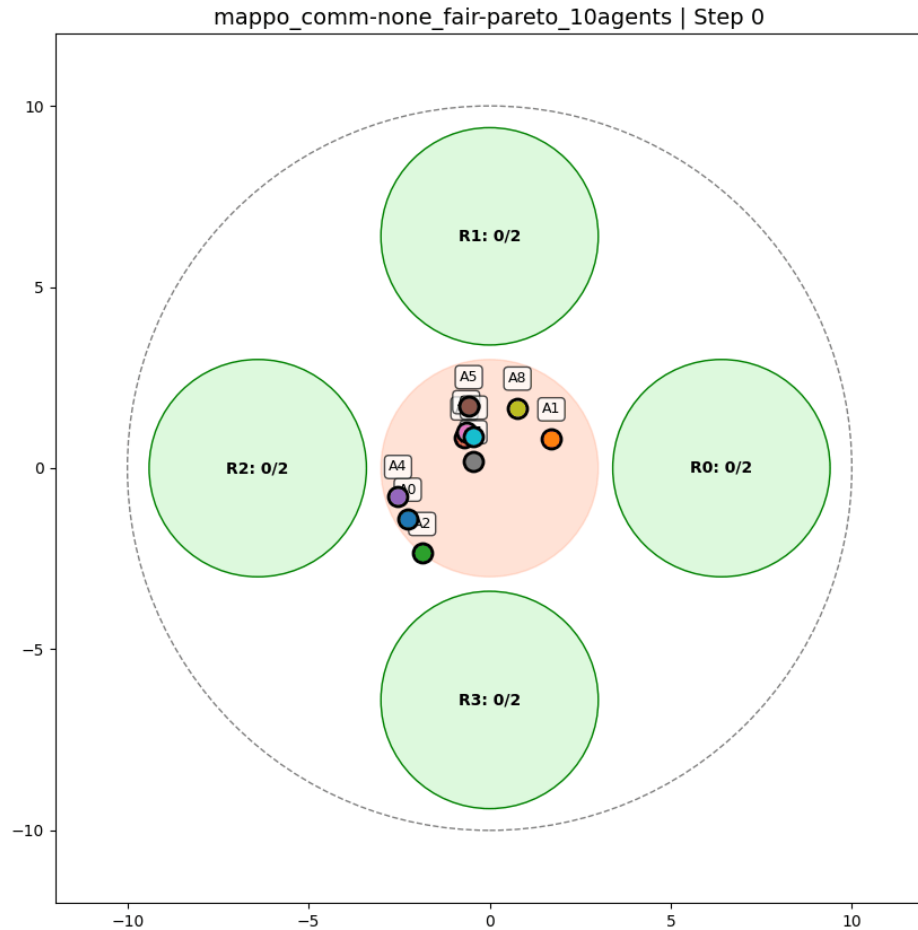
Key Results: Efficiency-Fairness Trade-Off



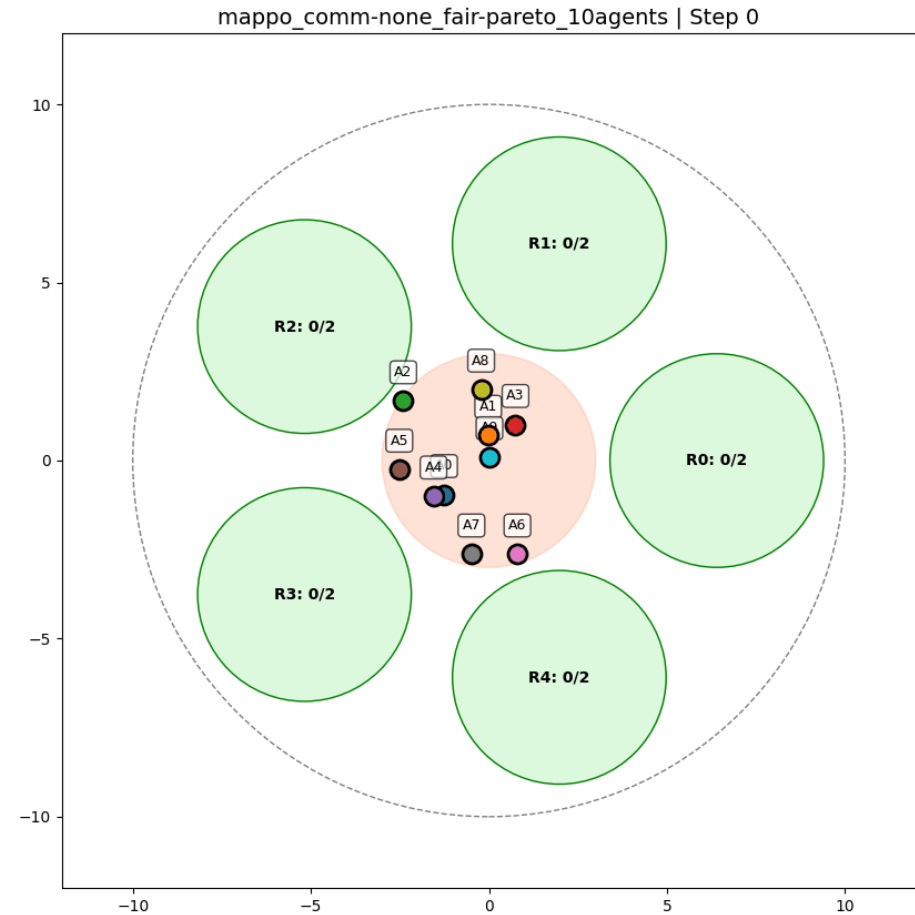
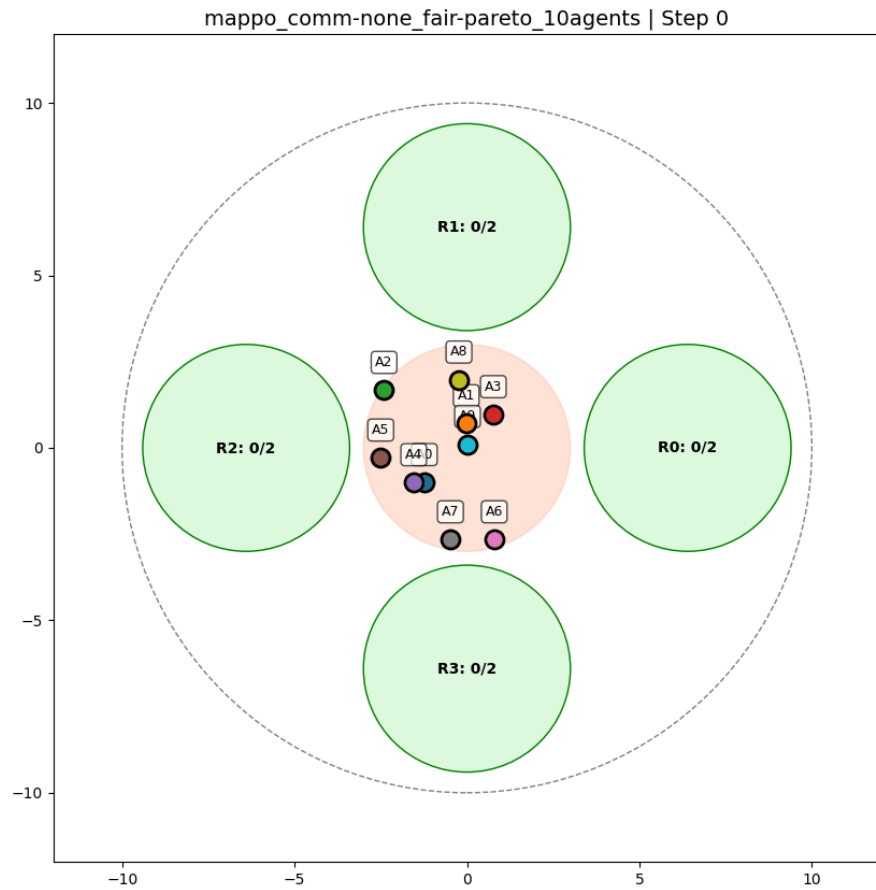
Qualitative Behavior – Alpha 0



Qualitative Behavior – Alpha 0.5



Qualitative Behavior – Alpha 1



Limitations & Future Work

Limitations:

- short CPU-based training budget
- fairness metrics changed only slightly
- large results may be treated as scalability validation if final reward update was not fully rerun

Future work:

- longer training
- stronger fairness objectives
- collision / room-switch / forced-exit analysis
- constrained optimization instead of scalarization

Thank you for your
attention!

